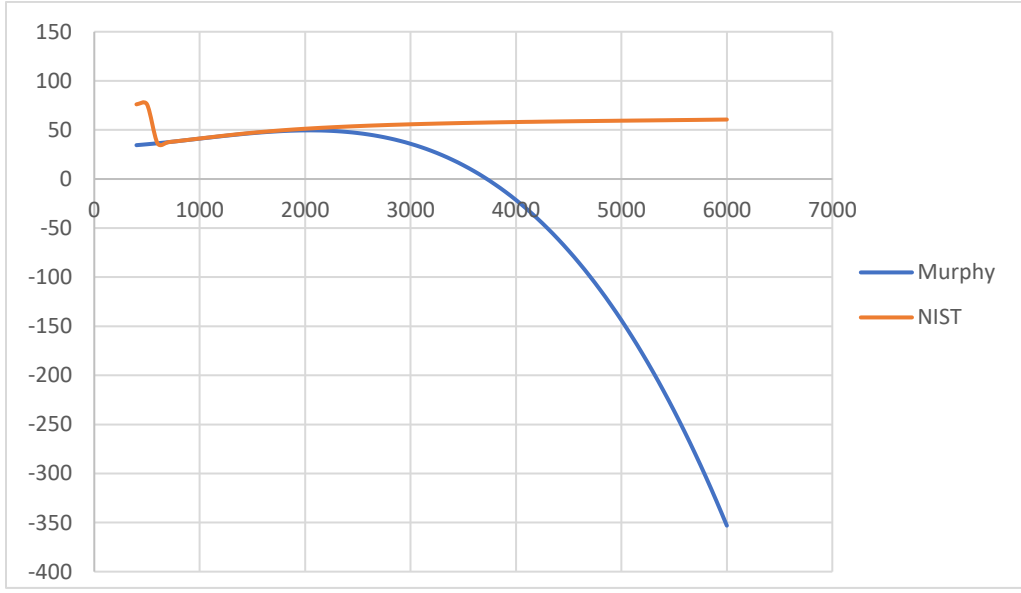
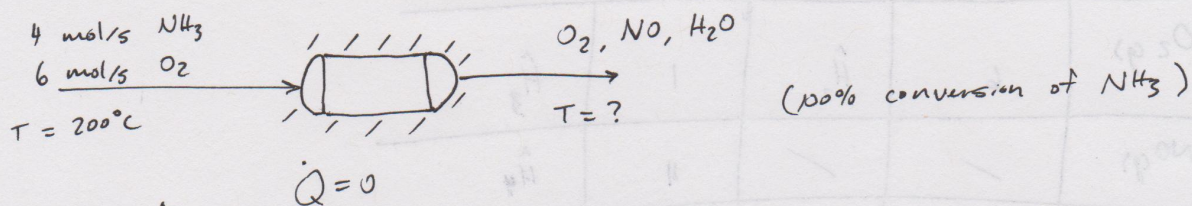
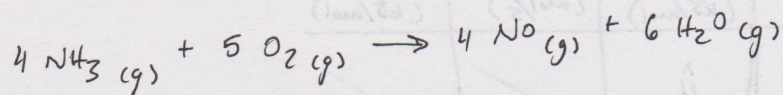


60
60

Extent of Rxn	1	mol/s					
Heat of Rxn	-905.9	kJ/mol					
Outlet Term	58.716	kJ/s					
$C_p^\circ = A + B \cdot t + C \cdot t^2 + D \cdot t^3 + E/t^2$							
Shomate Equation Coefficients:							
*NIST Chemistry WebBook	O2 (g)			NO (g)		H2O (g)	
Temp Range (K)	100-700	700 - 2000	2000 - 6000	298-1200	1200-6000	500-1700	1700-6000
A	31.32234	30.03235	20.91111	23.83491	35.99169	30.092	41.96426
B	-20.23531	8.772972	10.72071	12.58878	0.95717	6.832514	8.622053
C	57.86644	-3.988133	-2.020498	-1.139011	-0.148032	6.793435	-1.49978
D	-36.50624	0.788313	0.146449	-1.497459	0.009974	-2.53448	0.098119
E	-0.007374	-0.741599	9.245722	0.214194	-3.004088	0.082139	-11.15764
Eval @ Lower Bound: (.29815 K)	8.90319925	23.82279223	53.83858033	6.934477224	46.30250116	9.055126899	88.11016144
Eval @ Upper Bound:	21.40329725	70.49967417	71.9494217	36.05496552	92.50487075	66.81439012	128.0236894
Difference	12.500098	46.67688194	18.11084136	29.12048829	46.20236959	57.75926322	39.91352797
Energy Balance	-3.33955E-13						
Outlet Temp (K/1000):	2.473237027						
Temp (K)	2473.237						
Only 1 solution because this function only has 1 root							



[1] Oxidation of Ammonia:



Species	$\Delta \hat{H}_f^\circ$ (kJ/mol)	$\Delta \hat{H}_c^\circ$ (kJ/mol)
$\text{NH}_3 (\text{g})$	-46.15	-316.8
$\text{O}_2 (\text{g})$	0	0
$\text{NO} (\text{g})$	90.25	-90.2
$\text{H}_2\text{O} (\text{g})$	-241.83	0

$$\begin{aligned} \Delta \hat{H}_r^\circ &= \sum_i \nu_i \Delta \hat{H}_{f,i}^\circ = - \sum_i \nu_i \Delta \hat{H}_{c,i}^\circ \\ &= [4(90.25) + 6(-241.83) - 4(-46.15) - 5(0)] \frac{\text{kJ}}{\text{mol}} \\ \Delta \hat{H}_r^\circ &= -905.38 \frac{\text{kJ}}{\text{mol}} \end{aligned}$$

Material Balance:

$$\dot{n}_{\text{NH}_3, \text{out}} = \dot{n}_{\text{NH}_3, \text{in}} - 4 \dot{\xi} \Rightarrow \dot{\xi} = \frac{1}{4} \dot{n}_{\text{NH}_3, \text{in}} = 1 \text{ mol/s}$$

$$\dot{n}_{\text{O}_2, \text{out}} = \dot{n}_{\text{O}_2, \text{in}} - 5 \dot{\xi} \Rightarrow 6 - 5(1) = 1 \text{ mol/s}$$

$$\dot{n}_{\text{NO}, \text{out}} = 4 \dot{\xi} = 4 \text{ mol/s}$$

$$\dot{n}_{\text{H}_2\text{O}, \text{out}} = 6 \dot{\xi} = 6 \text{ mol/s}$$

Input / Output Table:

Species	$\dot{n}_{in}$ (mol/s)	$\hat{H}_{in}$ (kJ/mol)	$\dot{n}_{out}$ (mol/s)	$\hat{H}_{out}$ (kJ/mol)
$NH_3(g)$	4	$\hat{H}_1$	/	/
$O_2(g)$	6	$\hat{H}_2$	1	$\hat{H}_3$
$NO(g)$	/	/	4	$\hat{H}_4$
$H_2O(g)$	/	/	6	$\hat{H}_5$

$$\hat{H}_1 = \int_{298 K}^{473 K} (C_p)_{NH_3(g)} dT = 6.714 \frac{kJ}{mol}$$

$$\hat{H}_2 = \int_{298 K}^{473 K} (C_p)_{O_2(g)} dT = 5.31 \frac{kJ}{mol}$$

Using Murphy Table B.17

$$\hat{H}_3 = f(T), \quad \hat{H}_4 = f(T), \quad \hat{H}_5 = f(T) \Rightarrow \int_{298}^T C_p$$

Energy Balance:

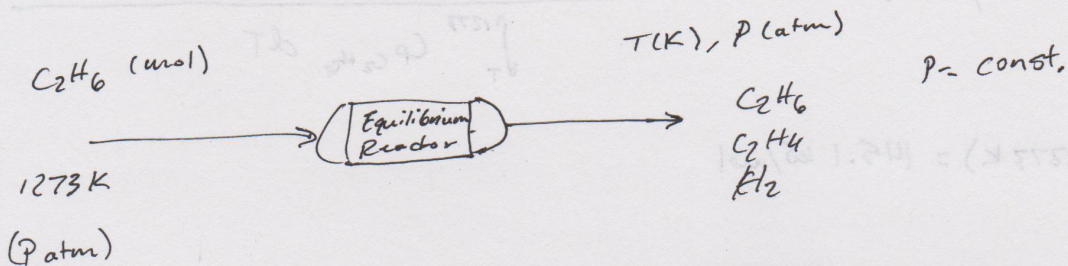
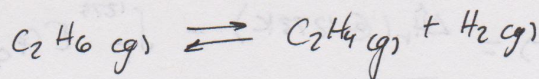
$$\dot{Q} = 0, \quad \Delta H = 0$$

$$0 = \sum_r \dot{n}_r \hat{H}_r + \sum_e \dot{n}_e \hat{H}_e - \sum_i \dot{n}_i \hat{H}_i$$

$$0 = (1 \frac{mol}{s}) (-90.5.9 \frac{kJ}{mol}) + (1 \frac{mol}{s}) (\hat{H}_3) + (4 \frac{mol}{s}) (\hat{H}_4) + (6 \frac{mol}{s}) (\hat{H}_5) - [(4 \frac{mol}{s}) (6.714 \frac{kJ}{mol}) + (6 \frac{mol}{s}) (5.31 \frac{kJ}{mol})]$$

Moving to excel:

[2]



From experiments:

$$K_p = \frac{x_{\text{C}_2\text{H}_4} x_{\text{H}_2}}{x_{\text{C}_2\text{H}_6}} P(\text{atm}) = K_0 e^{-E/T}$$

$$E = 17000 \text{ K}$$

$$K_0 = 7.28 \times 10^6 \text{ atm}$$

[a.] Prove that $f_{\text{C}_2\text{H}_6} = f = \sqrt{\frac{K_p}{P + K_p}}$

Material Balance

$$n_{\text{C}_2\text{H}_6, \text{out}} = n_{\text{C}_2\text{H}_6, \text{in}} - \frac{1}{2} \Rightarrow 1 \text{ mol} - \frac{1}{2} \therefore f = \frac{\frac{1}{2}}{n_{\text{C}_2\text{H}_6, \text{in}}}$$

$$n_{\text{C}_2\text{H}_4, \text{out}} = n_{\text{C}_2\text{H}_4, \text{in}} + \frac{1}{2} \quad x_{\text{C}_2\text{H}_6} = \frac{1-f}{1+f}$$

$$n_{\text{H}_2, \text{out}} = n_{\text{H}_2, \text{in}} + \frac{1}{2} \quad x_{\text{C}_2\text{H}_4} = \frac{f}{1+f}$$

$$n_{\text{total}, \text{out}} = 1 + \frac{1}{2} \quad x_{\text{H}_2} = \frac{f}{1+f}$$

$$K_p = \frac{\left(\frac{f}{1+f}\right) \left(\frac{f}{1+f}\right) P}{\left(\frac{1-f}{1+f}\right)} = \frac{f^2 P}{(1+f)(1-f)} = \frac{f^2 P}{1-f^2}$$

$$(1-f^2) K_p = f^2 P$$

$$K_p f^2 (P + K_p) \Rightarrow f = \sqrt{\frac{K_p}{P + K_p}} \quad P = 1 \text{ atm}$$

$$[6] \quad f = \frac{1}{1+\phi(T)}, \quad \phi(T) = \frac{\Delta \hat{H}_r(@1273K) - \int_T^{1273} [C_{p_{C_2H_4}} + C_{p_{H_2}}] dT}{\int_T^{1273} C_{p_{C_2H_4}} dT} \quad [5]$$

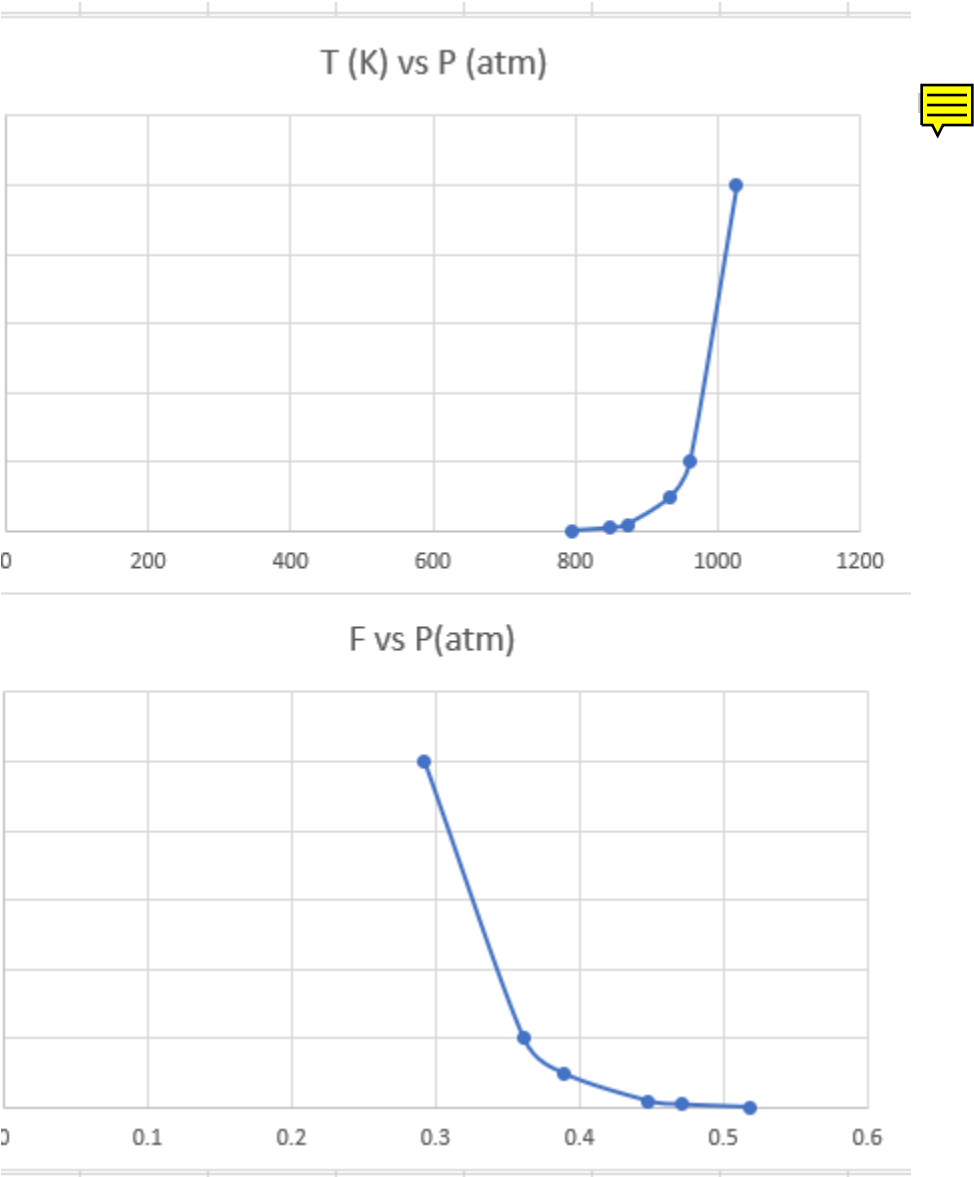
$$\Delta \hat{H}_r(@1273K) = 145.1 \text{ kJ/mol}$$

$$\psi(T) = \frac{1}{1+\phi(T)} - \sqrt{\frac{K_p}{P+K_p}} = 0$$

Excel: use solver to set $f = \sqrt{\frac{K_p}{P+K_p}}, \quad f = \frac{1}{1+\phi(T)}$
equal by changing temperature

P(atm)	T (K)	f	Kp(T)	f(Kp,P)	phi(T)	f(phi)			
0.01	794.2455	0.518946636	0.003685612	0.518946636	0.92698	0.518946636	Heat of Rx	145100	kJ/mol
0.05	847.7257	0.470576088	0.014221279	0.470576088	1.125055	0.470575962	Ko	7280000	atm
0.1	872.3451	0.447539271	0.025045548	0.447539271	1.234441	0.447539274	epsilon	17000	K
0.5	933.0312	0.388650387	0.088962228	0.388650387	1.573006	0.388650417			
1	960.5133	0.360983444	0.149833738	0.360983444	1.770208	0.360983684			
5	1026.366	0.292105172	0.46642508	0.292105172	2.423431	0.292104614			

Solver was used to set the to fraction selectivity functions equal by changing temperature



Comment Summary

Page 6

1. Nice work. Just remember to show the names for the axis